

Deliverable TP2 database

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November 2025

The project consist to implement an ETL pipeline between PostgreSQL, Neo4j, and a FastAPI service.

A collaborative filtering model based on graphs is an appropriate recommendation strategy for this configuration.

Neo4j's Graph Data Science library's Node Similarity, PageRank, and Community Detection graph algorithms can be used to find similar customers and suggest products by examining the relationships between customers and products (via PLACED and CONTAINS edges).

The following changes would be made to the project to make it production-ready:

- integrating unit tests, error management, and logging into the ETL and API layers.
- Using environment variables (with support for .env) for credentials.
- Docker health checks and CI/CD (GitHub Actions → DockerHub) automation.
- For scalability and resilience, the stack is being deployed on Kubernetes.
- Using Prometheus + Grafana to implement monitoring and alerting.

By doing this, the prototype would become a reliable, secure, and maintainable data pipeline that is prepared for actual enterprise use.